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$$T(n) = aT(n/b) + f(n)$$

$$C_{crit} = \log_b(a)$$

Handout 9
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6.046J/18.410J: Introduction to Algorithms
Professors Michel Goemans and Piotr Indyk

0, 0, 2

Master Theorem Worksheet

This is a worksheet to help you master solving recurrence relations using the Master Theorem. For each recurrence, either give the asymptotic solution using the Master Theorem (state which case), or else state that the Master Theorem doesn't apply. You should be able to go through these 25 recurrences in 10 minutes.

Problem 1-1. $T(n) = 3T(n/2) + n^2$

$$f(n) = n^2$$

$$C_{crit} = \log_2(3)$$

$$C > C_{crit}$$

$\therefore$ Case 3

$$T(n) = \Theta(n^2)$$

Problem 1-2. $T(n) = 7T(n/2) + n^2$

$$f(n) = n^2$$

$$C = 2$$

$$C < C_{crit}$$

$$C_{crit} = \log_2(7)$$

$\therefore$ Case 1

$$T(n) = \Theta(n^{\log_2(7)})$$

Problem 1-3. $T(n) = 4T(n/2) + n^2$

$$f(n) = n^2$$

$$C = 2$$

$$C = C_{crit}$$

$\therefore$ Case 2

$$C_{crit} = \log_2(4) = 2$$

$$f(n) = \Theta(n^2 (\log n)^k) = n^2$$

$$k = 0$$

$$T(n) = \Theta(n^2 (\log n)^0)$$

Problem 1-4. $T(n) = 3T(n/4) + n \lg n$

$$f(n) = n \lg n$$

$$n \lg n = \Omega(n^1)$$

$\therefore$ Case 3

$$C_{crit} = \log_4(3) = 0.79$$

$$C = 1, C_{crit} = 0.79$$

$$T(n) = \Theta(n \lg n)$$

Problem 1-5. $T(n) = 4T(n/2) + \lg n$

$$C > C_{crit}$$

$$f(n) = \lg n$$

$$\lg n = O(n^{1.8})$$

$\therefore$ Case 1

$$C_{crit} = \log_2(4) = 2$$

$$C = 1.8 < C_{crit}$$

$$T(n) = \Theta(n^2)$$

Problem 1-6. $T(n) = T(n-1) + n$

$$f(n) = n$$

doesn't apply?

$$n^2 \log(n) \quad n^{2.001}$$

2

Handout 9: Master Theorem Worksheet

Problem 1-7. $T(n) = 4T(n/2) + n^2 \lg n$

$$f(n) = n^2 \lg n$$

$$n^2 \lg n = \Theta(n^2 \log(n)^k) = \Theta(n^2 \log n)$$

$$c_{crit} = \log_2(4) = 2$$

$k=1$
 $\therefore$ case 2

$$T(n) = \Theta(n^2 (\log n)^2)$$

Problem 1-8. $T(n) = 5T(n/2) + n^2 \lg n$

$$f(n) = n^2 \lg n$$

$$n^2 \lg n = \Theta(n \log_2(4.5))$$

$$c_{crit} = \log_2(5)$$

$$c = \log_2(4.5) > 2$$

$\therefore$ case 1

$$T(n) = \Theta(n^{\log_2(5)})$$

Problem 1-9. $T(n) = 3T(n/3) + n/\lg n$

$$f(n) = n/\lg n$$

$$\frac{n}{\lg n} = \Theta(n \log(n)^k)$$

$$c_{crit} = \log_3(3) = 1$$

No case applies

Problem 1-10. $T(n) = 2T(n/4) + c$

$$f(n) = c$$

$$c = O(n^{0.25})$$

$\therefore$ case 1

$$c_{crit} = \log_4(2) = 1/2$$

$$c = 0.25 < 1/2 = c_{crit}$$

$$T(n) = \Theta(n^{1/2})$$

Problem 1-11. $T(n) = T(n/4) + \lg n$

$$f(n) = \lg n$$

$$\lg n = \Theta(n^0 \log(n)^k)$$

$k=1$

$$c_{crit} = \log_4(1) = 0$$

$$\lg n = \Theta(\log n)$$

$\therefore$ case 2

$$T(n) = \Theta((\log n)^2)$$

Problem 1-12. $T(n) = T(n/2) + T(n/4) + n^2$

Doesn't Apply

Problem 1-13. $T(n) = 2T(n/4) + \lg n$

$$f(n) = \lg n$$

$$\lg n = O(n^{.4})$$

$\therefore$ case 1

$$c_{crit} = \log_4(2) = 1/2$$

$$c = .4 < 1/2 = c_{crit}$$

$$T(n) = \Theta(n^{1/2})$$

Problem 1-14. $T(n) = 3T(n/3) + n \lg n$

$$f(n) = n \lg n$$

$$n \lg n = \Theta(n \log(n)^1)$$

$k=1$

$$c_{crit} = \log_3(3) = 1$$

$\therefore$ case 2

$$T(n) = \Theta(n (\log n)^2)$$

Problem 1-15. $T(n) = 8T((n - \sqrt{n})/4) + n^2$

Doesn't Apply

Problem 1-16. $T(n) = 2T(n/4) + \sqrt{n}$

$$f(n) = \sqrt{n}$$

$$\sqrt{n} = \Theta(n^{1/2} \log(n)^0) \quad (\text{case 2})$$

$$k=0$$

$$c_{\text{crit}} = \log_4(2) = 1/2$$

$$T(n) = \Theta(n^{1/2} \log(n))$$

Problem 1-17. $T(n) = 2T(n/4) + n^{0.51}$

$$f(n) = n^{0.51}$$

$$n^{0.51} = \Omega(n^{0.505}) \quad c = 0.505$$

$$c_{\text{crit}} = 1/2$$

$\therefore$ case 3

$$T(n) = \Theta(n^{0.51})$$

Problem 1-18. $T(n) = 16T(n/4) + n!$

$$f(n) = n!$$

$$n! = \Omega(n^{10})$$

$$c_{\text{crit}} = \log_4(16) = 2$$

$\therefore$ case 3

$$T(n) = \Theta(n!)$$

Problem 1-19. $T(n) = 3T(n/2) + n$

$$f(n) = n$$

$$n = O(n^{\log_2(2.1)})$$

$$c_{\text{crit}} = \log_2(3)$$

$\therefore$ case 1

$$T(n) = \Theta(n^{\log_2(3)})$$

Problem 1-20. $T(n) = 4T(n/2) + cn$

$$f(n) = cn$$

$$cn = O(n^{1.5})$$

$$c_{\text{crit}} = \log_2(4) = 2$$

$$c = 1.5 < 2$$

$\therefore$ case 1

$$T(n) = \Theta(n^2)$$

Problem 1-21. $T(n) = 3T(n/3) + n/2$

$$f(n) = n/2 = \frac{1}{2}n$$

$$\frac{1}{2}n = \Theta(n \log(n)^0)$$

$$c_{\text{crit}} = \log_3(3) = 1$$

$k=0 \therefore$ case 2

$$T(n) = \Theta(n \log(n))$$

Problem 1-22. $T(n) = 4T(n/2) + n/\lg n$

$$f(n) = 1/\lg(n)$$

$$\frac{1}{\lg(n)} = O(n^{-1})$$

$$c_{\text{crit}} = \log_2(4) = 2$$

$\therefore$ case 1

$$T(n) = \Theta(n^2)$$

Problem 1-23. $T(n) = 7T(n/3) + n^2$

$$f(n) = n^2$$

$$n^2 = \Omega(n^{\log_3(7)})$$

$\therefore$ case 3

$$C_{\text{crit}} = \log_3(7)$$

$$C = \log_3(8) < 2$$

$$T(n) = \Theta(n^2)$$

Problem 1-24. $T(n) = 8T(n/3) + 2^n$

$$2^n$$

$$2^n = \Omega(n^{16})$$

$$C_{\text{crit}} = \log_3(8)$$

$$C = 10 \dots \text{case 3}$$

$$T(n) = \Theta(2^n)$$

Problem 1-25. $T(n) = 16T(n/4) + n$

$$f(n) = n$$

$$n = O(n^{1.5})$$

case 1

$$C_{\text{crit}} = \log_4(16) = 2$$

$$C = 1.5$$

$$T(n) = \Theta(n^2)$$